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科目：

1. Denote that X_n is the RV, i.e.

$$X_n = \begin{cases} 1, & \text{system has crashed in } n\text{th day} \\ 0, & \text{system hasn't crashed in } n\text{th day} \end{cases}$$

$$P_{X_n}(x) = \begin{cases} 5\%, & x=1 \\ 95\%, & x=0 \\ 0\%, & \text{otherwise} \end{cases} \quad (\text{Let } p=5\%)$$

Denote ~~S_n~~ that S_n is a RV, i.e.

$$S_n = X_1 + X_2 + \dots + X_n$$

$$E[S_n] = n E[X_1] = np, \quad \text{var}(S_n) = np(1-p)$$

$$(a) \quad 0 \leq S_n \leq 5 \Leftrightarrow \frac{-np}{\sqrt{np(1-p)}} \leq \frac{S_n - np}{\sqrt{np(1-p)}} \leq \frac{5 - np}{\sqrt{np(1-p)}}$$

$$P\left(\frac{-np}{\sqrt{np(1-p)}} \leq \frac{S_n - np}{\sqrt{np(1-p)}} \leq \frac{5 - np}{\sqrt{np(1-p)}}\right) \approx \Phi\left(\frac{5 - np}{\sqrt{np(1-p)}}\right) - \Phi\left(\frac{-np}{\sqrt{np(1-p)}}\right)$$

$$\begin{aligned} \text{Let } n=50, p=5\%. \quad &= \Phi\left(\frac{2.5}{\sqrt{2.5 \times 95\%}}\right) - \Phi\left(\frac{-2.5}{\sqrt{2.5 \times 95\%}}\right) \\ &= 2 \times \left(\Phi\left(\frac{2.5}{0.95}\right) - \frac{1}{2}\right) = 2 \times 0.448 = 0.896 \end{aligned}$$

$$(b) \quad P_{S_n}(s) = \begin{cases} \binom{n}{s} p^s (1-p)^{n-s}, & 0 \leq s \leq n \\ 0, & \text{otherwise} \end{cases}$$

$$\binom{n}{s} p^s (1-p)^{n-s} \approx e^{-(np)} \frac{(np)^s}{s!} = e^{-2.5} \cdot \frac{2.5^s}{s!}$$

(Let $n=50, p=5\%$)

$$\begin{aligned} P(0 \leq S_n \leq 5) &= P_{S_n}(0) + P_{S_n}(1) + P_{S_n}(2) + P_{S_n}(3) + P_{S_n}(4) + P_{S_n}(5) \\ &\approx e^{-2.5} \times \sum_{i=0}^5 \frac{2.5^i}{i!} = 0.958 \end{aligned}$$

2. $E[X_n] = 5$, $\text{var}(X_n) = 9$

$$X_1 + X_2 + \dots + X_n \leq 440 \Leftrightarrow \frac{X_1 + X_2 + \dots + X_n - nE[X]}{\sqrt{n \text{var}(X)}} \leq \frac{440 - nE[X]}{\sqrt{n \text{var}(X)}} = \frac{440 - 5n}{\sqrt{9n}}$$

$$(a) P(X_1 + X_2 + \dots + X_n \leq 440) = P\left(\frac{X_1 + X_2 + \dots + X_n - nE[X]}{\sqrt{n \text{var}(X)}} \leq \frac{440 - 5n}{\sqrt{9n}}\right) \approx \Phi\left(\frac{440 - 5n}{\sqrt{9n}}\right)$$

$$\text{Let } n=100, \Phi\left(\frac{440 - 5 \times 100}{\sqrt{9 \times 100}}\right) = \Phi\left(-\frac{61}{30}\right) = 0.021$$

$$(b) X_1 + X_2 + \dots + X_n \geq 200 + 5n \Leftrightarrow \frac{X_1 + X_2 + \dots + X_n - nE[X]}{\sqrt{n \text{var}(X)}} \geq \frac{200}{\sqrt{9n}}$$

$$P(X_1 + \dots + X_n \geq 200 + 5n) \geq P\left(\frac{X_1 + X_2 + \dots + X_n - nE[X]}{\sqrt{n \text{var}(X)}} \geq \frac{200}{\sqrt{9n}}\right) \approx 1 - \Phi\left(\frac{200}{3\sqrt{n}}\right)$$

$$1 - \Phi\left(\frac{200}{3\sqrt{n}}\right) \leq 0.05 \Leftrightarrow \Phi\left(\frac{200}{3\sqrt{n}}\right) \geq 0.95$$

As According to the table $\Phi(1.65) \approx 0.95$ (In real, $\Phi(1.65) \geq 0.95$)

$$\Rightarrow \frac{200}{3\sqrt{n}} \geq 1.65 \Leftrightarrow n \leq \left(\frac{200}{3 \times 1.65}\right)^2 \approx 1632$$

$$n_{\max} = \left\lfloor \left(\frac{200}{3 \times 1.65}\right)^2 \right\rfloor = 1632$$

$$(c) N \geq 220 \Leftrightarrow X_1 + X_2 + \dots + X_{219} \leq 1000$$

$$P(N \geq 220) = P(X_1 + X_2 + \dots + X_{219} \leq 1000)$$

$$\text{Same as part (a), } P(X_1 + X_2 + \dots + X_{219} \leq 1000) \approx \Phi\left(\frac{1000 - 5 \times 219}{\sqrt{9 \times 219}}\right)$$

$$= \Phi(-2.1398) = 1 - \Phi(2.1398) \approx 1 - 0.9838 = 0.0162$$

3. $E[X] = E[Y] = \frac{1}{2}$, $\text{var}(X) = \text{var}(Y) = \frac{1}{12}$

$$E[W] = \frac{1}{16} \times 16 \times (E[X] - E[Y]) = 0, \quad \text{var}(W) = \frac{1}{16^2} \times 16 \times (\text{var}(X) + \text{var}(Y)) = \frac{1}{96}$$

$$|W - E[W]| < 0.001 \Leftrightarrow \left| \frac{W - E[W]}{\sqrt{\text{var}(W)}} \right| \leq \frac{0.001}{\sqrt{\frac{1}{96}}} = 0.001 \sqrt{96}$$

$$\Rightarrow P(|W - E[W]| < 0.001) = P\left(-0.001 \sqrt{96} \leq \frac{W - E[W]}{\sqrt{\text{var}(W)}} \leq 0.001 \sqrt{96}\right) \approx \Phi(0.001 \sqrt{96}) - \Phi(-0.001 \sqrt{96})$$

$$= 2 \times \left(\Phi(0.001 \sqrt{96}) - \frac{1}{2}\right) = 0.0078$$

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4. Denote that X_n, Y_n are the RVs, i.e.

$$X_n = \begin{cases} 1, & \text{the } n\text{th mother is a smoker} \\ 0, & \text{otherwise} \end{cases}$$

$$Y_n = \begin{cases} 1, & \text{the } n\text{th baby } \overset{\text{looks}}{\text{is}} \text{ abnormal} \\ 0, & \text{otherwise} \end{cases}$$

~~Denote C_n is a RV, i.e.~~

~~$$C_n = X_n Y_n + (1 - X_n)(1 - Y_n)$$~~

~~Denote S_n is a RV, i.e.~~

~~$$S_n = C_1 + C_2 + \dots + C_n$$~~

~~if the conjecture is not plausible, then X_n, Y_n are independent~~The answer of the ~~conjecture~~ is correct $\Leftrightarrow X_n = Y_n \Leftrightarrow X_n Y_n + (1 - X_n)(1 - Y_n) = 1$
nthDenote $C_n = X_n Y_n + (1 - X_n)(1 - Y_n)$. Assume that X_n, Y_n are independentWithout loss of generality, $X_n = \begin{cases} 1, & n \in [1, 750] \\ 0, & n \in [751, 1500] \end{cases}$

$$C_n = \begin{cases} Y_n, & n = 1, 2, \dots, 750 \\ 1 - Y_n, & n = 751, \dots, 1500 \end{cases}$$

~~If the conjecture isn't plausible,~~~~Denote $S_n = C_1 + C_2 + \dots + C_n$, $E[S_n] =$~~ Denote $S = C_1 + C_2 + \dots + C_{1500}$, $E[S] = 750 E[Y] + 750 - 750 E[Y] = 750$

$$\text{var}(S) = 1500 \text{var}(Y) \leq 1500 \times \frac{1}{4} = 375$$

$$P(S \geq 909) = P\left(\frac{S - E[S]}{\sqrt{\text{var}(S)}} \geq \frac{909 - 750}{\sqrt{\text{var}(S)}}\right) = P\left(\frac{S - E[S]}{\sqrt{\text{var}(S)}} \geq \frac{159}{\sqrt{\text{var}(S)}}\right) \approx 1 - \Phi\left(\frac{159}{\sqrt{\text{var}(S)}}\right) \leq 1 - \Phi\left(\frac{159}{\sqrt{375}}\right)$$

$$\approx 1 - \Phi\left(\frac{53\sqrt{15}}{25}\right) \approx 0.$$

It means that the probability of $S \geq 909$ almost is 0Therefore, X_n, Y_n isn't independent $\Rightarrow$ the conjecture is plausible.